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HUJI Applicative Research Report — Agriculture & Food — Week 44, 2026

1 paper(s) · Agriculture & Food · Weekly · Week 44, 2026 · Generated October 26, 2026

■■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's digest highlights a significant advancement in AgriTech and Synthetic Biology from Professor Alexander Vainstein's lab. Researchers have pinpointed the genetic complex (MBW) and a specific repressor (EOBV) that precisely controls floral scent production in petunias. This foundational discovery provides a direct target for genetically engineering ornamental plants with custom-designed fragrances. The commercial implications are vast, promising innovative product lines across horticulture, high-end perfumery, and the broader consumer fragrance market.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Alexander Vainstein** — AgriTech, Synthetic Biology, FoodTech

AGRICULTURE & FOOD — RESEARCH HIGHLIGHTS

1. The MBW complex regulates floral volatiles in petunia: transcriptional repressor EOBV is a MYB component of the complex. 14/50

"Genetically engineer custom flower scents for high-value horticulture, perfumery, and consumer markets."

Main researcher: Alexander Vainstein

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ABOUT THIS RESEARCH

Researchers at Hebrew University of Jerusalem uncovered that the MBW complex, a transcriptional regulator, controls the production of floral volatiles in petunia through the repressor EOBV, a MYB protein. This discovery elucidates the genetic basis of scent biosynthesis and offers a target for manipulating flower fragrance.

COMMERCIAL ANGLE

By engineering the MBW complex or EOBV, companies can develop petunias and other ornamental plants with customized scents, opening markets in horticulture, perfumery, and consumer fragrances.

COMMERCIALISATION METRICS

Novelty 4/10

This research provides a incremental mechanistic characterization of well-understood MBW regulatory complexes in model plant systems, representing a standard academic contribution rather than a groundbreaking, platform-enabling, or high-IP-potential innovation.

Commercial Potential 3/10

The study is basic plant biology with limited immediate translation to a clear product or licensable technology; commercial potential is modest and indirect.

Market Size 2/10

The study focuses on a specific transcriptional regulator in petunia floral scent, a niche area with limited commercial application; the addressable market is small and not broadly industrially relevant.

Tech Readiness 2/10

This is fundamental plant molecular biology research focused on elucidating transcriptional regulatory mechanisms in a model organism. It provides scientific insight into gene function rather than developing a commercial product, platform, or industrially scalable application.

IP Strength 3/10

This paper describes fundamental biological mechanisms of gene regulation in model plants, which constitutes basic scientific discovery rather than a proprietary, commercially enforceable technology. While the insights could inform future metabolic engineering, the findings themselves lack the novelty and industrial utility required for strong patentability.

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